

Sprint Documentation

The core data layer is designed to manage complex aviation tracking through precise relationship definitions across four architectural layers: Controller, Service, Repository, and Entity.

City to Airport (One to Many)

A single municipality can govern multiple airport transit hubs. This is mapped using standard `@OneToMany` and `@ManyToOne` annotations, ensuring foreign key constraints are strictly maintained at the database level.

Passenger to Aircraft (Many to Many)

This complex relationship tracks detailed passenger manifests. It utilizes a dedicated join table to map associations, allowing the system to accurately query which aircraft a specific passenger has boarded, and conversely, which passengers were on a specific flight.

Pagination Implementation

To prevent payload bloating and ensure optimal query performance on heavy data endpoints, we integrated Spring Data's `Pageable` interface.

CLI Client Integration

The Java terminal application acts as an independent, interactive consumer of the Spring Boot API.

- **The CLI constructs and dispatches targeted HTTP GET requests to the API based on user terminal input.**
- **Incoming JSON payloads are parsed and mapped to local Data Transfer Objects (DTOs) to ensure type safety before terminal rendering.**
- **The client supports specialized analytical queries, including fetching city specific airport rosters, individual passenger transit histories, and global fleet tracking records.**

Security & Workflow Constraints

To guarantee comprehensive stack mastery and eliminate knowledge silos, the repository adhered to strict cross functional development constraints.